Gold tailings database: parameters' variability and correlations

Nicolas Tasso, Nicolas A. Labanda, Kuntan Chang, Riccardo Fanni
Mine Tailings, WSP, Perth, Australia, nicolas.tasso@wsp.com

ABSTRACT: A comprehensive database was developed to analyse laboratory test results on tailings samples collected from gold mines (gold tailings), focusing on the variability of key physical and mechanical parameters. The study explores material classification, permeability, strength, and critical state parameters, providing valuable insights into the geotechnical behaviour of gold tailings. Results are compared with existing correlations from the literature, including both tailings and natural soils, to assess their reliability and applicability to gold tailings. This comparison improves our understanding of this material's behaviour and supports evidence-based engineering practices and future research methodologies.

KEYWORDS: Tailings, Gold, Geotechnics, Critical State, Characterisation, Database

1 INTRODUCTION

Mine operations generate roughly 13 billion tons of tailings every year, and the stability of the impoundments that contain this waste has emerged as a major global risk (Hudson-Edwards et al., 2024). As gold is one of the most extensively mined commodities, laboratory characterisation of its tailings has attracted considerable attention. However, significant gaps remain in understanding their behaviour.

Smith et al. (2019) evaluated an extensive tailings database including gold tailings. Most of the data was from specimens prepared in a loose state adopting the moist-tamping technique, sheared under triaxial compression conditions to derive their critical-state-line (CSL) parameters. Torres-Cruz & Santamarina (2020) compiled 25 published CSLs and linked the intercept void ratio and slope to simple index parameters such as the minimum void ratio. Shipton and Coop (2015) defined the transitional behaviour for sands with plastic and non-plastic fines (i.e. a non-unique CSLs dependent on fabric) feature consistently found across tailings materials. Their study highlights the need to account for fabric effects in liquefaction assessments, a controversial argument according to critical state soil mechanics (CSSM) theory. Complementing these efforts, Fanni et al (2025) executed the first constant-shear-drained hollow-cylinder tests on sandy–silt gold tailings, investigating the effect of anisotropy under stress paths representative of below-slope loading.

Despite these remarkable advances, the literature remains dominated by single-site campaigns, leaving designers without a quantitative envelope of parameter bounds, classifications, or critical-state parameters across different commodities groups.

The present paper consolidates a gold tailings database compiled throughout eight years of WSP Perth laboratory testing, complemented by data from the literature. The tailings were sourced from both oxidised-weathered and fresh rock tailings generated ore bodies. Statistical techniques are applied to quantify the variability of gradation, Atterberg limits, and strength parameters, with the intent to identify systematic trends associated with sample preparation methods and parameters from standard classification testing. Critical-state parameters (i.e. λ_e and Γ) are then benchmarked against established correlations for natural soils to evaluate their applicability to gold tailings. The resulting empirical relationships and variability envelopes may be useful to develop site-investigation plans, scoping and pre-feasibility level dam designs, constitutive-model calibration when laboratory testing are not readily available and both probabilistic and/or deformation-based slope-stability assessments. By providing the dataset and analytical workflow, the study supports evidence-based design and lays a foundation for future research into the behaviour of gold tailings.

2 DATABASE DESCRIPTION

The database was developed using the WSP Perth laboratory tailings data from more than twenty gold mine projects, with the number of tests summarized in Table 1. The results discussed in this paper correspond to the following tests: particle size distribution including hydrometry (PSD), Atterberg limits, X-ray diffraction (XRD), flexible wall permeability (FWP), monotonic direct simple shear (MDSS), isotropically consolidated undrained triaxial test (CIU), K_0 consolidated undrained triaxial test (CK0U), anisotropically consolidated undrained triaxial test (CAU), isotropically consolidated drained triaxial test (CID), K_0 consolidated drained test (CK0D), and anisotropically consolidated drained triaxial test (CAD).

Table 1. Summary of tests considered on the analysed database.

Test	Quantity	Test	Quantity
PSD	99	TX CK0U	38
Atterberg limits	36	TX CAU	14
XRD	19	TX CID	102
FWP	26	TX CK0D	1
MDSS	101	TX CAD	2
TX CIU	73		

3 GRAIN SIZES, PLASTICITY AND PERMEABILITY

This section analyses the PSD, plasticity (i.e. liquid limit, LL, and plasticity index, PI) and XRD testing to characterise the typical properties of the gold tailings.

Figure 1 presents the PSD for the studied dataset together with the corresponding specific gravity (Gs). Gold tailings samples are shown to have a fines content varying from 25% to 100% depending on various aspects like ore body features, processing plant, etc. The Gs exhibited a broad range, typically from 2.6 to 3.6 depending on the associated mineralogy. The higher Gs values are associated with higher concentration of hematite, while the lower values correspond to higher concentration of kaolin (oxidised ore body).

Figure 2 shows the plasticity chart, where most of the samples can be classified as low plasticity clay, while a minor fraction falls into the intermediate to high plasticity clay category.

Figure 3 shows the XRD results illustrating the mineralogical composition of the tailings samples. Most samples are predominantly composed of quartz, muscovite and hematite, with occasional traces of clinochlore, and gypsum. One sample represents a special case, exhibiting a dominant

kaolinite content due to an oxidation process in part of the ore body, becoming the only high-plasticity sample in the dataset.

Figure 4 correlates the permeability (k) with the void ratio (e) and the fines content (FC), following a classical geotechnical approach (Terzaghi et al, 1996). A good correlation for practical purposes is observed, where samples with higher fraction of fines are grouped on the range of permeabilities between $1x10^{-7}$ to $1x10^{-9}\ m/s$, while the coarser tailings exhibit upper bound permeability, i.e. up to $1x10^{-5}\ m/s$.

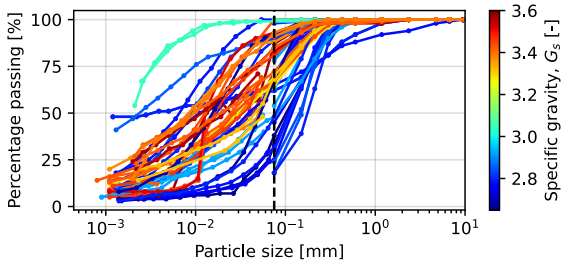

Figure 1. Particle size distribution (PSD).

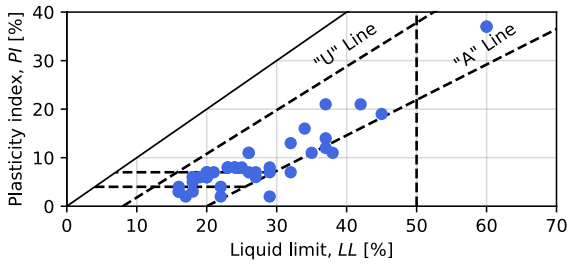

Figure 2. Plasticity chart.

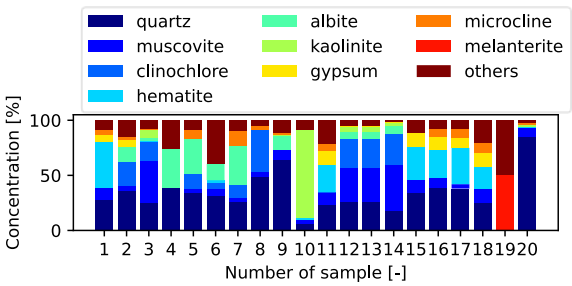

Figure 3. Components concentration on X-ray diffraction (XRD) results.

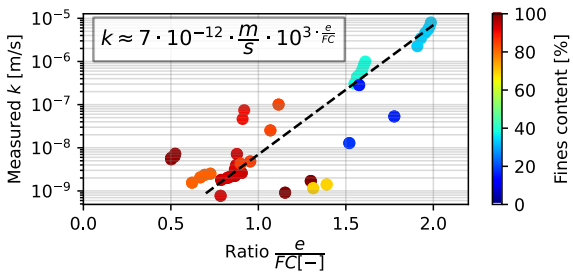

Figure 4. Correlation between k, void ratio and fines content.

4 DRAINED AND UNDRAINED STRENGTH

Here, the triaxial tests results for both isotropic and anisotropic consolidation conditions are compiled to evaluate the drained and undrained shear strength properties. Low plasticity sandy silt and silty sand tailings were remoulded using the moist tamping preparation technique to enable obtaining specimens of loose enough consistency to not localise during shearing. The higher plasticity samples where instead prepared adopting the slurry methods while a smaller subset were extruded from piston tubes samples obtained in the field and tested as intact.

Figure 5 shows the mobilised friction angle (ϕ_{mob}) versus axial strain (ε_a) for all 254 tests, with a colour bar showing the initial state parameter (ψ_0) of each sample. Figure 6 shows the histograms of the maximum friction angle (ϕ_{max}) and critical friction angle (ϕ_c), fitted with a Laplace and Normal probability density distribution functions. Results are separated into dense samples $\psi < -0.05$ shown in red, and loose samples $\psi \geq -0.05$ shown in green. The ψ cut-off of -0.05 was used as this value is typically used in literature as the contractive/dilative boundary (Reid, 2023). ϕ_c values were calculated using only loose samples. Undrained triaxial testing with mean effective stress (p') at critical state less than 50 kPa were removed from the data interpretation to avoid spurred statistical evaluation of frictional strength due to limitation on the devices used to carry out the testing to measure accurate stresses at low stresses and the high influence on membrane correction on this parameter.

The mean ϕ_c is ~34.3°, with a standard deviation of ~2.8°, and 90% of the samples exhibit values higher than 32°. This value is similar to the critical friction angle reported by Jefferies and Been (2016) for tailings.

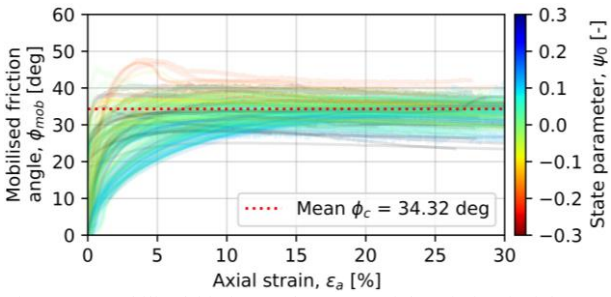

Figure 5. Mobilised friction angle versus axial strain in triaxial tests.

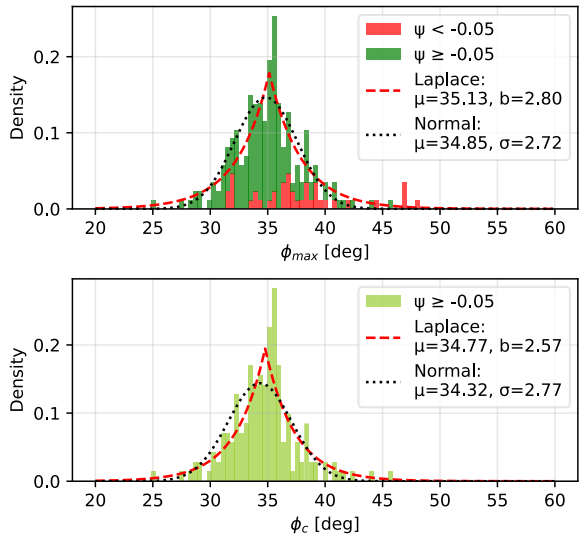

Figure 6. Probabilistic distribution of maximum and critical state friction angles.

Figure 7 shows the evolution of the ϕ_c as a function of PI, published in the classical Terzaghi et al (1996) book. The gold tailings dataset using the results for obtained for contractive samples extends the literature trends for natural soils into the range of low-plasticity materials.

While the data suggest a tendency toward higher friction angles at lower PI, the authors acknowledge that the dataset is not sufficiently large to support a definitive conclusion, and therefore these results should be interpreted with caution.

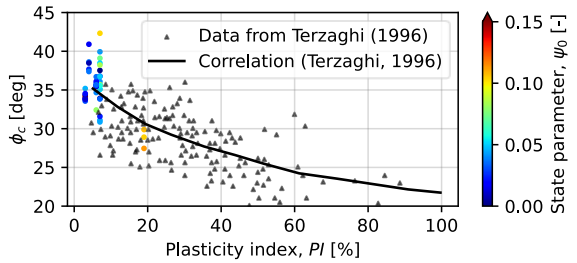

Figure 7. Critical friction angle versus plasticity index correlation (Terzaghi et al, 1996).

Figure 8 shows the undrained shear strength for triaxial tests $s_{u,TX}$ (105 tests) and for MDSS $s_{u,MDSS}$ (101 tests). To allow distinguishing between testing carried out on loose and dense prepared specimens, the data are coloured based on the ψ_0 calculated from their correspondent CSL. As in MDSS the principal stresses are unknown, a K_0 value of 0.5 was assumed to estimate the ψ_0.

Lower bound lines for the whole data set and for K_0 consolidated triaxial tests are added, suggesting a minimum undrained shear strength ratio $USR_{all} = 0.23$, and $USR_{K0} = 0.30$, respectively. On the other hand, the MDSS data set exhibits a lower bound $USR_{MDSS} = 0.17$, leading to an anisotropic strength ratio $USR_{K0}/USR_{MDSS} = 0.57$, and $USR_{all}/USR_{MDSS} = 0.74$, which falls within typical ranges (Ladd, 1991).

Figure 9 presents the histogram and lognormal fits for the undrained strength ratios obtained from triaxial and MDSS tests. The mode value for the triaxial test is 0.27, whereas for MDSS tests it is 0.20.

Figure 10 compares the undrained shear strength ratio presented in Ladd (1991) for natural normally consolidated soils as white-filled markers, with the presented database shown in colour-filled markers. To allow proper comparison with Ladd's data, in this graph only the data exhibiting contractive behaviour are plotted (i.e., ψ > 0) and consolidated under K_0 conditions.

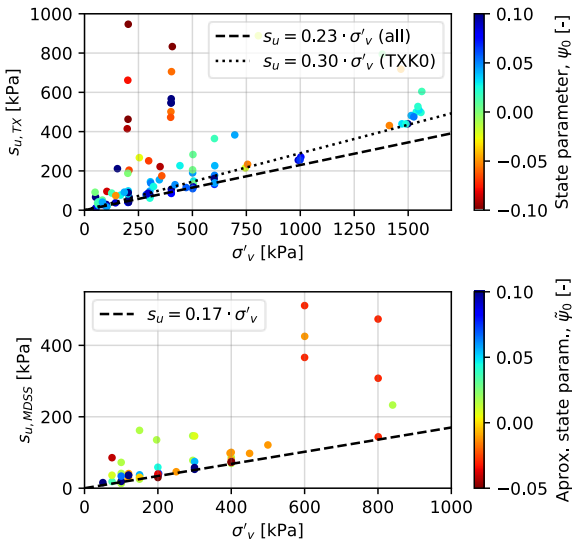

Figure 8. Undrained shear strength from triaxial and MDSS.

Our triaxial and MDSS data exhibit strength within the Ladd (1991) dataset for both K_0 consolidated triaxial and MDSS. A greater scatter is observed compared to the literature trends, suggesting that existing correlations proposed by Ladd are not directly applicable to this dataset, especially for MDSS. This discrepancy may be attributed to several factors: the more angular particle morphology of tailings relative to natural soils, which increases the friction angle in silty and/or clayey materials and promotes sand-like behaviour; as well as the specimen preparation methods commonly employed in tailings testing, such as moist tamping and slurry consolidation. Nevertheless, the authors acknowledge that the dataset in this case is reduced, and this plot may be revisited in future works.

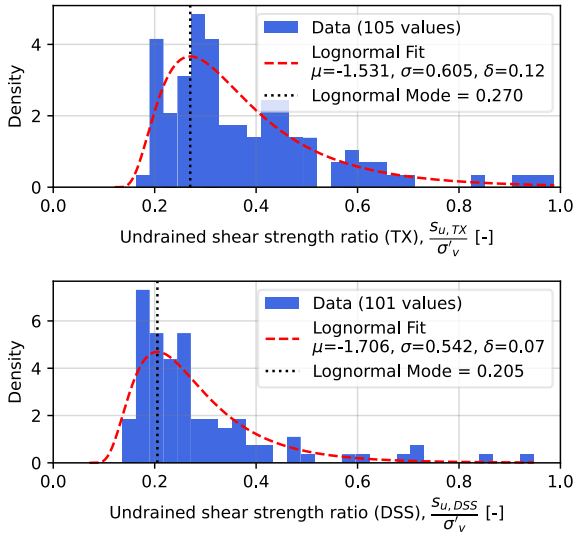

Figure 9. Undrained shear strength ratio distribution for triaxial and MDSS tests.

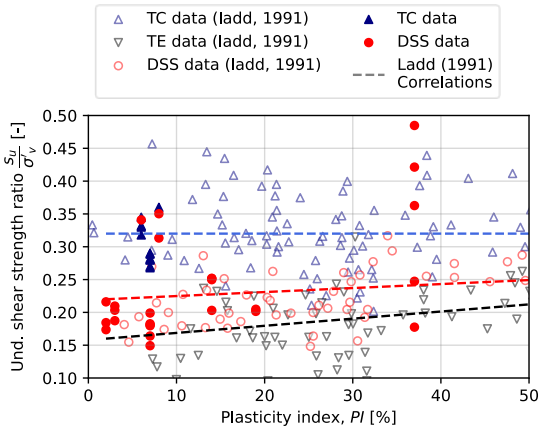

Figure 10. Undrained shear strength ratio and comparison with Ladd (1991).

5 TAILINGS STIFFNESS AND K0 VALUES

This section reinterprets the triaxial test dataset to estimate the drained stiffness parameter at 50% of the peak strength, normalized to a reference mean effective pressure of 100 kPa, denoted as E_{50}^{ref}. In parallel, the consolidation stages of the MDSS tests are used to derive the oedometric stiffness, E_{oed}^{ref}, also referenced to 100 kPa. For these calculations, a Jambu coefficient $m = 0.5$ is assumed. The authors acknowledge that this assumption may lead to a systematic underestimation of stiffness and should be reassessed in future studies.

Figure 11 presents the histogram of E_{50}^{ref} values for specimens prepared by moist tamping and slurry deposition, along with a lognormal distribution fit for each preparation method. The mean ψ_0 of the moist tamped specimens was 0.033 while that of the slurry prepared specimens was -0.027.

The results indicate that slurry-deposited specimens consistently exhibit higher stiffness values compared to those

prepared by moist tamping. Specifically, the mode of the slurry-deposited group falls within the 7,000–8,000 kPa range, whereas the moist-tamped specimens exhibit a modal bin between 4,000 and 5,000 kPa. When analysing the oedometric stiffness E_{oed}^{ref} of the gold tailings MDSS consolidation stage dataset, the most likely values fall in the modal bin between 2000-3000 kPa, being the same order of the moist tamped E_{50}^{ref}.

It is noted that the dataset of the tailings database was produced from loosely prepared specimens with different mean ψ_0 values for the moist-tamped and slurry prepared specimens. Hence, it is reasonable to assume that the stiffness calculated for the two subpopulations of data may have likely been affected by the obtained ψ_0, wherein higher mean ψ_0 are likely to exhibit lower overall stiffness compared to lower mean ψ_0. Nevertheless, it is also reasonable to assume that differences in fabric between the moist-tamped and slurry prepared specimens is likely another contributor role. The effect of moisture content and initial density on the stress-strain behaviour of gold tailings was clearly observed by Reid et al (2023), wherein specimens prepared using a dense-tamping procedure exhibited significant higher stiffness and peak undrained shear strengths compared to specimen prepared at higher moisture content and lower initial density, indicating the clear effect of fabric on tailings response.

The authors note that estimating stiffness parameters at strain levels relevant to TSF performance remains particularly challenging, owing to the difficulty of retrieving intact specimens and the pronounced sensitivity of stiffness to sample preparation method. Consequently, the stiffness values reported herein should be taken with caution, in particular if a deterministic approach is used to infer deformations.

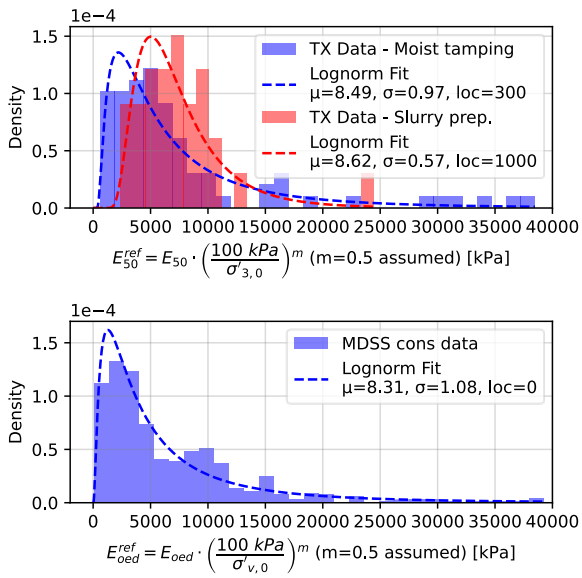

Figure 11. Tailings stiffness E_{50}^{ref} and E_{oed}^{ref}obtained from triaxial and MDSS tests.

Figure 12 presents the distribution of final K_0 values obtained at the end of the consolidation stage for CK0 tests, categorised by sample preparation method: moist tamped, slurry deposited, and push-tube extruded specimens. The histogram reveals that most values cluster between 0.42 and 0.57, with two notable outliers near 0.65. The outliers are from CK_0 testing carried out on a specimen extruded from a tube sample sheared at p' = 300 kPa and a slurry prepared specimen sheared from p' = 50 kPa. Comparison with estimates from the Jaky (1944) equation indicated that the empirical relation consistently underpredicts the observed values. This indicates that, while the Jaky formulation yields a conservative estimate, it may inadequately represent the influence of depositional fabric and reconstitution method on the consolidation behaviour of the tailings materials.

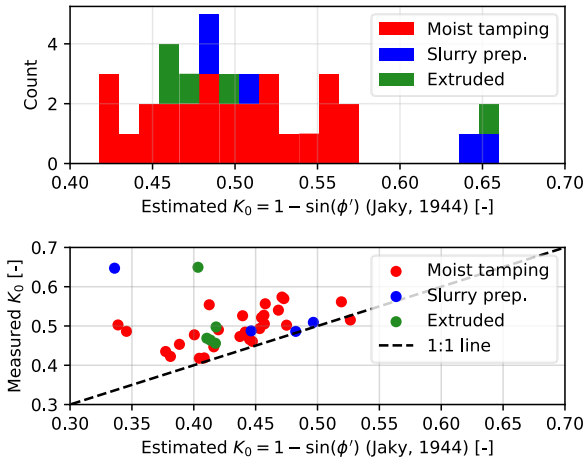

Figure 12. Comparison between K_0 obtained from lab tests and Jaky equation (1944).

6 CRITICAL STATE PROPERTIES

This section presents a dataset of 216 tests undertaken to establish the critical state locus (CSL) in the compression plane (e-p') with e normalized by the Gs. Figure 13 grouped the results by sample preparation method and includes the leached ore samples as dashed lines. The moist tamped samples exhibit a consistent trend, whereas slurry prepared samples show greater scatter.

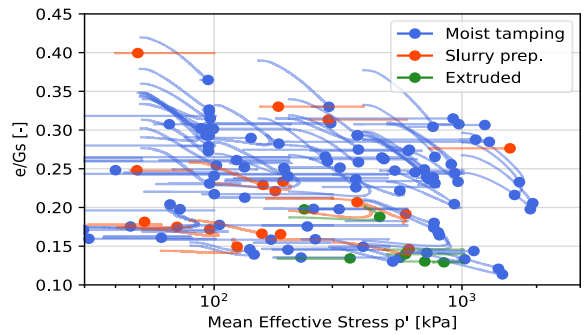

Figure 13. Critical state locus from triaxial tests moist tamped, slurry prepared, and push-tube extruded samples.

Figure 14 shows the interpretated CSL for all samples, with the fines content represented in the colour bar. Most cases (44) exhibit CSL represented by a semi-log expression ($e_c = \Gamma - \lambda_e \ln(p')$), characteristic of predominantly clay-like materials. In contrast, four cases display a power-law CSL expression ($e_c = A - B \cdot (p'/p_{ref})^{-n}$), corresponding to sandier tailings materials. This behaviour is typically observed in the literature.

Figure 15 presents the histogram and fit functions for the critical state parameters Γ (CSL intercept) and λ_e (CSL slope, a proxy of material compressibility), based on the dataset exhibiting semi-log CSL expression. The mean value of Γ lies between 1.0 and 1.1 with a normal probability distribution. The mean value of λ_e falls within 0.03-0.05, with a lognormal probability distribution.

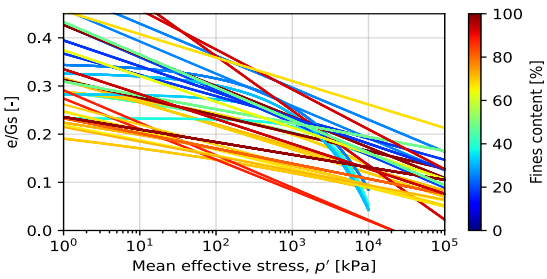

Figure 14. Critical state locus for studied dataset.

Figure 16 shows the CSL slope λ_e for the whole dataset, including the data exhibiting power-law CSLs, as a function of the clay-size fraction. For the power-law CSLs, λ_e is calculated from the derivative of the CSL at various stress ranges, i.e. from 0 kPa to 2MPa. The results indicate that samples with clay content larger than 12% exhibits typically a semi-log CSL, whereas those with a lower clay content tends to display a power-law expression. Moreover, within the materials presenting power-law CSL expression, multiple lines corresponding to the λ_e calculated for high (red dashed lines) and low stresses (bark blue dashed lines) are included to illustrate the evolution of λ_e with stresses. While previous findings can serve as a general guideline, it should be noted that the dataset is limited for samples with higher clay contents. Accordingly, this plot should be revisited and reassessed as additional data become available.

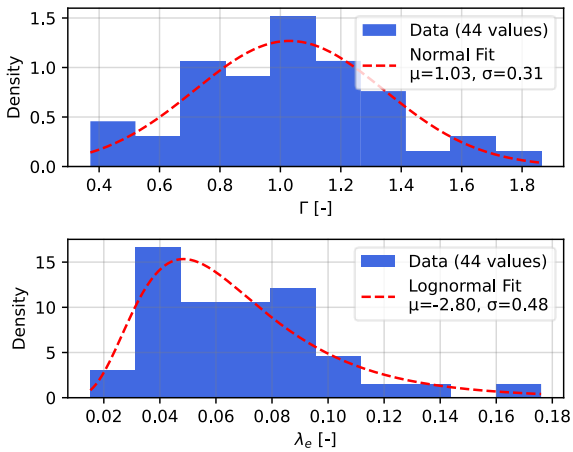

Figure 15. CSL parameters histograms and lognormal fitting.

Figure 17 shows the correlation between the liquid limit LL and the void ratio normalized by the specific gravity Γ/G_s, including fines content in the colour bar. Generally, a good correlation is observed: material with higher LL exhibit higher Γ/G_s. In addition, a good correlation between Γ and λ_e is also clear, following a linear trend. These two plots provide a useful tool for computing and comparing new test results.

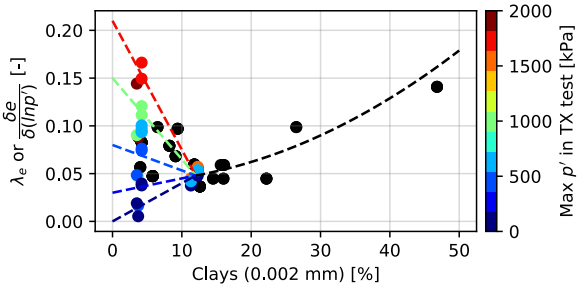

Figure 16. Influence of clays on the CSL shape.

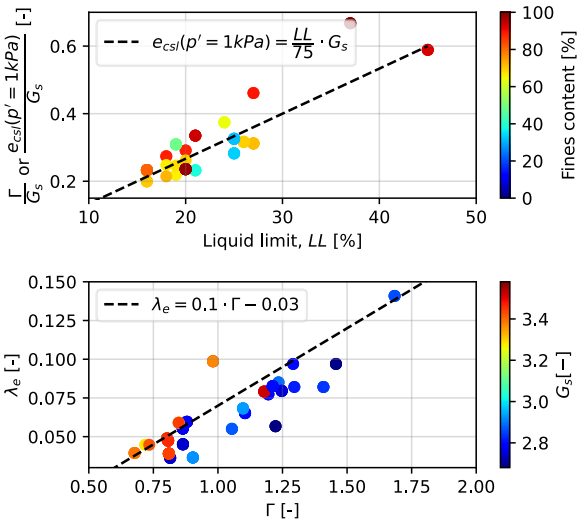

Figure 17. Correlations between liquid limit (LL) and Γ, and correlation between Γ and λ_e.

7 STATE PARAMETER INFLUENCE ON STRENGTH AND BRITTLENESS

Following Jefferies and Been (2016) approach, the peak shear strength and the residual shear strength as a function of the state parameter is presented in Figure 18 for all triaxial tests presented before. For the peak strength, the present cloud of points appears to be in similar range with the literature data for natural soils, although more scatted in high state parameter ranges. The coloured bars which are expressing tests sheared from K_0 consolidation shows that higher initial stress ratio (i.e. lower K_0) tends to mobilise higher peak shear strengths, hence plotting on the higher bound of the dataset.

On the other hand, the second plot compares the residual shear strength against the steady-state strength line, NorSand prediction and the best practice trend for $\lambda_{10} \approx 0.11$, with the colour bar representing the fines content of the sample. As expected, triaxial tests generally follow the steady-state line with some other dots located toward the NorSand line for materials with state parameters lower than 0.1. These are tests that typically exhibit quasi-steady state (QSS) behaviour (Alarcon-Guzman et al., 1999).

Figure 19 presents the dilatancy expressed as maximum friction angle minus critical friction angle versus the state parameter for our dataset, alongside the relationship reported by Been (2015) for natural soils and tailings. Our results do not reproduce the literature trend for contractive samples, instead yielding a scattered distribution of points with no discernible correlation. On dilative samples, a more consistent trend towards higher dilatancies is observed, where those results are compared with Been (2015) correlation, and an upper and lower given by the sample preparation method.

Figure 20 plots the obtained brittleness index as a function of the relation between the state parameter and the compressibility ψ_0/λ_e, comparing the current dataset with relationships proposed by Smith et al (2019). In this plot, colour bar represents the K_0 value at the consolidation stage.

Here, the more isotopically consolidated samples exhibit larger brittleness index, ranging from I_B less than 0.1 for isotropically consolidated samples, to moderate to high I_B exceeding 0.5 for K_0 consolidated samples. The data fall within the upper and lower boundaries defined by Smith et al (2019), for both isotropically and anistropically consolidated samples.

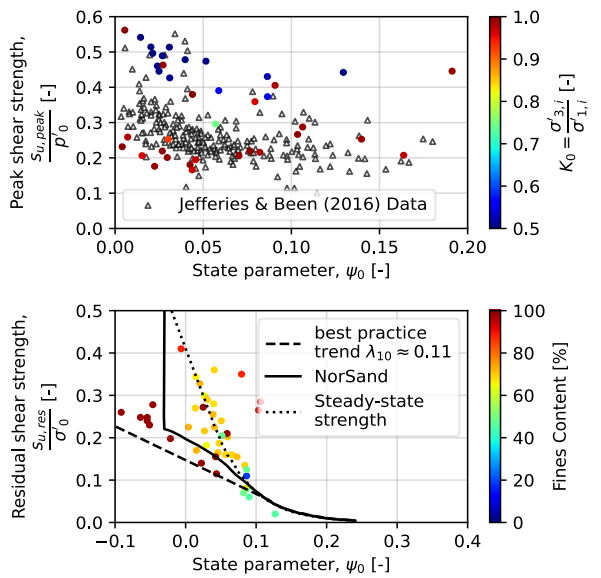

Figure 18. Peak and residual shear strength as a function of the state parameter. Comparison with Jefferies and Been (2016).

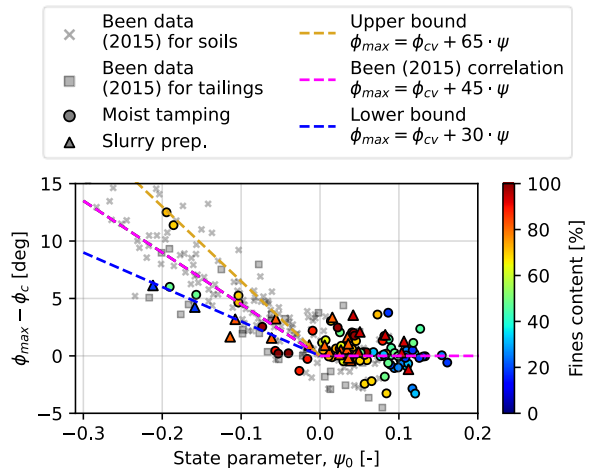

Figure 19. Critical state friction angle as a function of the state parameter, including comparison with Been (2015) data.

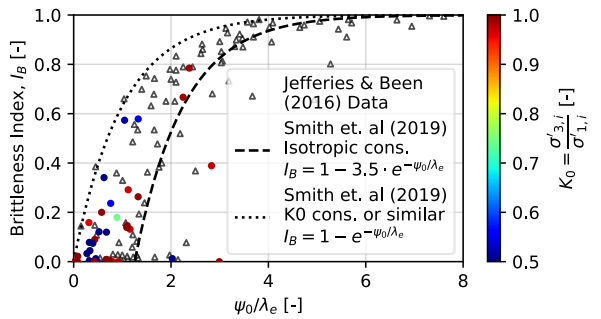

Figure 20. Brittleness index as a function of state parameter, including relationships proposed by Smith et al (2019).

8 CONCLUSIONS

The variability observed in the compiled dataset underscores the complexity of characterising gold tailings, where differences in geological origin, mineralogy, and processing history result in broad parameter ranges for strength, stiffness, compressibility, and critical state properties. Such diversity, compounded by the influence of particle morphology and depositional fabric, limits the reliability of empirical correlations derived from natural soils and general tailings databases, and cautions against the use of single "representative" values for design.

Mineralogical analyses confirmed wide compositional variability, with quartz, hematite, and muscovite common, and occasional kaolinite-rich samples linked to higher plasticity. Despite consistently high fines content, plasticity was generally low. Triaxial testing indicated relatively high critical state friction angles, with weak correlations to fines or plasticity, highlighting the dominant role of angularity and interlocking in shear resistance. Stiffness parameters were strongly influenced by preparation method, with slurry samples stiffer and more homogeneous than moist-tamped ones, and K_0 values generally above Jaky's prediction (1944) but still preparation-dependent. Critical state lines were consistent within preparation subgroups but more scattered in slurry samples. The dataset validated NorSand trend for residual shear strength for dilative materials, at the same time peak shear strengths showed to follow a similar scatter that the results presented in Jefferies and Been (2016). Moreover, dilatancy parameters showed to follow and increasing trend for contractive samples (Been, 2015), and brittleness index showed to be within the boundaries proposed by Smith et al. (2019).

Within this context, the presented database should be regarded as a reference for framing expected ranges and identifying potential outliers, not as a prescriptive design tool. For any given tailings storage facility, targeted laboratory testing on representative material remains indispensable to capture the combined effects of local geology, mineralogy, and deposition conditions. Only by integrating such site-specific data with broader database trends can practitioners develop robust, evidence-based designs that are resilient to the inherent heterogeneity of gold tailings.

9 REFERENCES

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